

SACRAMENTO MCCLELLAN, CA, USA

WMO: 724836

Lat: 38.667N

Lon: 121.400W

Elev: 24

StdP: 101.04

Time Zone: -8.00 (NAP)

Period: 94-19

WBAN: 23208

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -0.2	(c) 1.1	(d) -7.4	(e) 2.0	(f) 11.0	(g) -4.8	(h) 2.5	(i) 12.6	(j) 12.3	(k) 13.3	(l) 11.1	(m) 12.3	(n) 1.2	(o) 320	(p) 0.470

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 17.5	(c) 38.8	(d) 20.7	(e) 37.3	(f) 19.9	(g) 35.2	(h) 19.1	(i) 21.9	(j) 36.0	(k) 21.0	(l) 34.3	(m) 20.2	(n) 33.1	(o) 3.1	(p) 290

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.4	25.4	16.1	11.5	24.6	15.1	10.8	23.1	63.6	36.1	60.5	34.5	57.8	33.1	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9.6	8.0	6.9	-2.9	41.2	2.1	1.1	-4.5	42.1	-5.7	42.7	-6.9	43.4	-8.4	44.2
			-3.1	23.3	2.1	1.5	-4.6	24.4	-5.8	25.2	-7.0	26.0	-8.5	27.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.1	9.3	10.9	13.2	15.9	19.4	23.5	25.7	25.3	23.1	17.9	12.4	8.3
	DBStd	6.75	2.67	2.50	2.77	3.35	3.34	3.51	2.79	2.86	2.95	2.83	3.11	2.91
	HDD10.0	147	43	17	5	1	0	0	0	0	0	0	12	68
	HDD18.3	1295	280	207	160	87	26	3	0	0	2	41	179	310
	CDD10.0	2741	22	43	104	178	291	405	486	476	392	246	83	17
	CDD18.3	848	0	0	1	14	58	158	227	217	144	28	1	0
	CDH23.3	10774	0	0	26	247	788	2063	2867	2653	1683	428	19	0
(14) Wind	WSAvg	3.2	2.9	3.2	3.2	3.4	3.7	3.7	3.7	3.5	3.1	2.8	2.5	2.8
	PrecAvg	420	85	74	64	27	10	4	1	2	7	23	58	62
	PrecMax	850	232	218	181	107	53	32	20	16	71	174	188	159
	PrecMin	159	4	5	0	0	0	0	0	0	0	0	0	0
	PrecStd	159	64	58	47	28	12	6	4	4	14	31	52	43
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.0	21.3	26.3	30.8	36.0	39.9	40.3	41.1	38.4	32.9	25.9	18.0
		MCWB	12.0	14.3	15.5	17.2	19.4	21.5	21.2	21.4	20.5	17.8	15.0	13.0
	2%	DB	17.3	19.7	23.6	28.0	32.9	37.8	38.5	38.0	35.4	29.8	22.9	16.3
		MCWB	12.3	13.7	14.6	16.0	17.9	20.2	20.6	20.1	18.8	17.0	14.2	12.2
	5%	DB	15.2	17.7	22.1	26.5	30.9	35.2	37.2	36.4	33.7	27.6	20.9	15.0
		MCWB	11.6	12.2	14.1	15.7	17.7	19.2	19.8	19.4	18.2	15.1	13.7	11.3
	10%	DB	13.9	16.3	19.9	23.9	27.8	32.9	34.9	34.1	32.2	25.8	18.0	13.8
		MCWB	11.1	12.3	13.3	14.8	16.2	18.4	19.3	18.9	17.8	14.5	12.6	10.7
	0.4%	WB	16.1	16.3	17.6	19.3	20.8	22.9	23.1	22.6	22.3	19.0	16.8	15.5
		MCDB	17.4	18.5	23.0	26.8	31.4	37.1	37.1	37.7	34.3	30.8	21.7	17.1
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	14.0	15.0	16.0	17.7	19.3	21.5	21.8	21.4	20.6	17.7	15.8	14.0
		MCDB	15.3	17.3	21.3	24.9	29.9	35.2	36.0	35.8	32.8	27.4	20.2	15.2
	5%	WB	12.7	13.9	15.0	16.5	18.2	20.4	20.8	20.4	19.5	16.6	14.8	12.8
		MCDB	14.2	16.7	20.0	24.2	28.7	33.1	34.4	34.0	31.5	24.9	18.7	14.2
	10%	WB	11.8	12.8	13.9	15.5	17.0	19.3	20.0	19.6	18.5	15.7	13.8	11.5
		MCDB	13.1	15.4	18.4	22.6	26.2	31.2	33.2	32.6	29.6	23.0	17.1	13.0
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	10.0	11.4	13.1	14.1	16.5	17.5	17.2	16.0	14.7	11.1	9.4
		MCDBR	11.3	12.7	15.4	17.1	18.1	19.2	19.3	19.2	18.6	17.6	15.1	10.7
		MCWBR	7.1	7.3	7.8	7.2	6.5	6.6	6.7	6.5	6.4	6.8	7.1	6.6
	5% WB	MCDBR	7.2	10.2	13.0	15.1	16.7	18.3	18.4	18.0	16.9	14.7	11.3	6.5
		MCWBR	5.4	6.6	7.1	7.0	6.9	6.7	7.1	6.4	6.2	6.0	6.3	4.8
(40) Clear-Sky Solar Irradiance	taub		0.323	0.326	0.346	0.356	0.370	0.364	0.364	0.347	0.355	0.349	0.331	0.336
	taud		2.507	2.507	2.436	2.402	2.384	2.410	2.406	2.443	2.431	2.449	2.486	2.457
	Ebn at Noon		855	903	914	921	910	911	908	918	892	861	836	806
	Edh at Noon		78	89	105	115	120	117	116	109	104	92	79	77
(44) All-Sky Solar Radiation	RadAvg		2.17	3.19	4.56	6.05	7.37	8.25	8.22	7.37	5.96	4.20	2.68	1.93
	RadStd		0.40	0.41	0.49	0.53	0.46	0.34	0.24	0.23	0.21	0.28	0.20	0.33

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (46 neighbors)	+0.40	N/A	-1.08	N/A	N/A	N/A	N/A	-116	+138	+42

Nomenclature: See separate page